

# ALI AKARMA

Trustworthy AI & Agentic Systems Researcher

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[www.aliakarma.codes](http://www.aliakarma.codes) | [GitHub](https://github.com/aliakarma) | [Google Scholar](https://scholar.google.com/citations?user=...) | [LinkedIn](https://www.linkedin.com/in/aliakarma)

## RESEARCH PROFILE

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Undergraduate AI researcher specializing in trustworthy agentic systems, adversarial robustness, safety-constrained reinforcement learning, and human oversight for autonomous AI. For Fall 2027, I aim to extend this work to trustworthy multimodal and speech-enabled AI, including robust audio-language models, voice agents, and safety evaluation of multimodal agents.

## RESEARCH INTERESTS

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Trustworthy and safe agentic AI; adversarial robustness and security of LLM-based autonomous systems; safety-constrained multi-agent reinforcement learning; human oversight and value-aligned control; trustworthy multimodal and speech-enabled AI; robust audio-language models and voice agents; multimodal AI safety evaluation.

## EDUCATION

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**B.S. Information Technology** — Islamic University of Madinah, Saudi Arabia 2023 – May 2027

- GPA: 4.31 / 5.00 — *Merit-based fully funded scholarship (competitive international selection)*
- Relevant coursework: Machine Learning, Algorithms & Data Structures, Probability & Statistics, Database Systems, Software Engineering
- Thesis interest: Trustworthy multimodal agent orchestration under adversarial conditions

## RESEARCH EXPERIENCE

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**Research Intern** — King Fahd University of Petroleum and Minerals (KFUPM) Jun 2026 -- Present

- Investigating verification of autonomous AI decision-making using reinforcement learning and multi-agent systems, with emphasis on safety, reliability, distributional shift, and adversarial conditions.
- Developing evaluation protocols for trustworthy autonomous-agent policies under distribution shift and adversarial scenarios.

**Undergraduate AI Researcher** — Islamic University of Madinah Jan 2025 -- Present

- Designed constrained multi-agent reinforcement learning architectures using C-Dec-POMDP and Lagrangian CTDE for safety critical prediction, achieving 81.5 reward with a 2.3% safety-violation rate across six baselines.
- Evaluated adversarial prompt-injection and jailbreak failure modes in autonomous LLM-agent pipelines; developed LangChain defence middleware with three detection modes and four policy strategies.
- Developed a governance-constrained agentic AI framework incorporating human oversight for safety-critical autonomous decision-making; first author, ICETAS 2026.
- Built 20+ reproducible open-source research systems with automated tests and Zenodo-archived datasets across agentic AI, safety-critical decision-making, healthcare, and smart infrastructure.

## RESEARCH PROJECTS & FUNDING

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**King Salman Center for Disability Research (KSCDR)** — Research Group No. KSRG-2026-321

- Siddiqui, M. S., Syed, T. A., & Akarma, A. (2026). "ADAPT: An Agentic AI Framework for People with Disabilities and Neurodivergence". *Journal of Disability Research*, 5(2), e20260830.

DOI: [10.57197/JDR-2026-0830](https://doi.org/10.57197/JDR-2026-0830) | GitHub: [github.com/aliakarma/ADAPT](https://github.com/aliakarma/ADAPT)

- Syed, T. A., Siddiqui, M. S., Akarma, A., & Formisano, A. (2026). "FedAgent-Chain: A Secure Federated and Agentic AI Framework for Multilingual Disability-Inclusive Employment in AI Cities". *Smart Cities*, 9(7), 106.

DOI: [10.3390/smartcities9070106](https://doi.org/10.3390/smartcities9070106) | GitHub: [github.com/aliakarma/fedagent-chain](https://github.com/aliakarma/fedagent-chain)

## SELECTED RESEARCH PUBLICATIONS

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**Google Scholar:** 71 citations | h-index: 5 | i10-index: 2 (September 2026).

### First-Author Papers

- **Akarma, A.**, Syed, T. A., Khan, M., & Ahmad, A. (2026). Privacy Leakage in Federated Learning: Gradient-Based Client Identity Inference and Defenses for Inertial Sensing in Vehicular Edge Networks. **104th VTC Fall Boston** [arXiv:2609.02971](https://arxiv.org/abs/2609.02971)
- **Akarma, A.**, Syed, T.A., Jan, S., Muneer, H., Jilani, A.K. "Governance-Constrained Agentic AI: Blockchain-Enforced Human Oversight for Safety-Critical Wildfire Monitoring," **10th ICETAS 2026**. [arXiv:2604.04265](https://arxiv.org/abs/2604.04265)
- **Akarma, A.**, Syed, T.A., Jilani, A.K., Jan, S., Muneer, H., Khan, M.A., Yu, C. "Agents for Agents: An Interrogator-Based Secure Framework for Autonomous Internet of Underwater Things," **10th ICETAS 2026**. [arXiv:2604.04262](https://arxiv.org/abs/2604.04262)
- **Akarma, A.**, Syed, T.A., Hameed, D., Muneer, H. "Improving Disability Support and Institutional Performance With AI," Book Chapter, **IGI Global**. [preprints202608.1878](https://preprints202608.1878)

### Journal Articles

- Syed TA, **Akarma A.**, Alatif A, Naqash MT, Alqurashi A. "Agentic AI-enhanced digital twins for Smart City civil infrastructure: A secure, autonomous and auditable management framework" **PLOS ONE**. [10.1371/journal.pone.0353610](https://doi.org/10.1371/journal.pone.0353610) (Co-First Author).
- Jan, S., **Akarma, A.**, Syed, T.A., Muhammad, M. A., & Kamal, S. "EAGF: a four-pillar ethical AI governance framework for trustworthy cybersecurity in 5G renewable energy IoT systems". **Scientific Reports** 2026. [10.1038/s41598-026-63383-5](https://doi.org/10.1038/s41598-026-63383-5).
- Syed, T. A., **Akarma, A.**, Naqash, M. T., Hameed, D., Kamal, S., & Formisano, A. "Agentic AI for Climate-Resilient Cities: A PRISMA-Guided Review and Digital Twin Framework," **MDPI Sustainability**. [10.3390/su18178917](https://doi.org/10.3390/su18178917)
- Syed, T. A., Alshahrani, A., **Akarma, A.**, Khan, S., Nauman, M., Lee, I. E., ... & Ullah, A. "FinNutriAgent (FNA): An Agentic AI for Budget and Nutrition Planning," **ETASR**, 2026. [10.48084/etasr.15640](https://doi.org/10.48084/etasr.15640)

### Conference Papers

- Syed, T. A., Alshahrani, A., Ullah, A., **Akarma, A.**, Khan, S., Nauman, M., & Jan, S. "FinAgent: Agentic AI for Personal Finance and Nutrition," **IEEE ICCA 2025**. [10.1109/ICCA66035.2025.11430760](https://doi.org/10.1109/ICCA66035.2025.11430760)
- Jan, S., Razzaqi, H. A., **Akarma, A.**, & Belgaum, M. R "Blockchain-Monitored Agentic AI Architecture for Trusted Perception-Reasoning-Action Pipelines," **IEEE ICCA 2025**. [10.1109/ICCA66035.2025.11430865](https://doi.org/10.1109/ICCA66035.2025.11430865)

## OPEN-SOURCE RESEARCH SOFTWARE

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All repositories include reproducible experimental pipelines and Zenodo-archived datasets.

- **Trustguard** — Constrained multi-agent reinforcement learning framework for trustworthy runtime permission governance using MAPPO and Lagrangian safety constraints. ([Repository](#))
- **Secure Agent Runtime** — Multimodal security framework for autonomous LLM agents, covering prompt injection, provenance, and text, image, audio, and video threats. ([Repository](#))
- **SAS-Shield** — Safety-assurance framework for agentic AI combining conformal testing, formal verification, control-barrier constraints, adversarial evaluation, and cryptographic auditability. ([Repository](#))

Additional research software and reproducibility artifacts: [github.com/aliakarma](https://github.com/aliakarma)

## TECHNICAL SKILLS

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**AI/ML:** PyTorch, Hugging Face Transformers, scikit-learn, NumPy

**Agentic AI & LLMs:** LLM orchestration, autonomous agents, prompt-injection defense, adversarial robustness

**Reinforcement Learning:** MAPPO, CTDE, C-Dec-POMDP, Lagrangian constrained optimization

**Trustworthy AI:** safety evaluation, failure-mode analysis, human oversight, policy-aligned control, auditability

**Research Engineering:** Python, Git, Linux, Jupyter, pytest, LaTeX, Zenodo

## AWARDS & CERTIFICATIONS

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- **Merit-Based Fully Funded Scholarship** — Islamic University of Madinah (competitive international selection, 2023)
- **Certificate of Research Excellence** — ICETAS 2026, for research contribution on agentic AI governance
- **Certificate of Appreciation** — ICBDT 2025, for presented research contributions

## LANGUAGES

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**Languages:** Urdu (native), English (professional proficiency), Arabic (professional proficiency)